
AWS GLOBAL INFRASTRUCTURE

(The Foundation of High Availability & Resilience)

This is one of the most important Cloud Practitioner topics.

If students truly understand this, they understand how AWS delivers reliability.

1 Why Global Infrastructure Matters

Before cloud:

- Companies built one data center.
- If it failed → everything failed.
- Scaling required buying more hardware.
- Disaster recovery was expensive.

AWS changed this model.

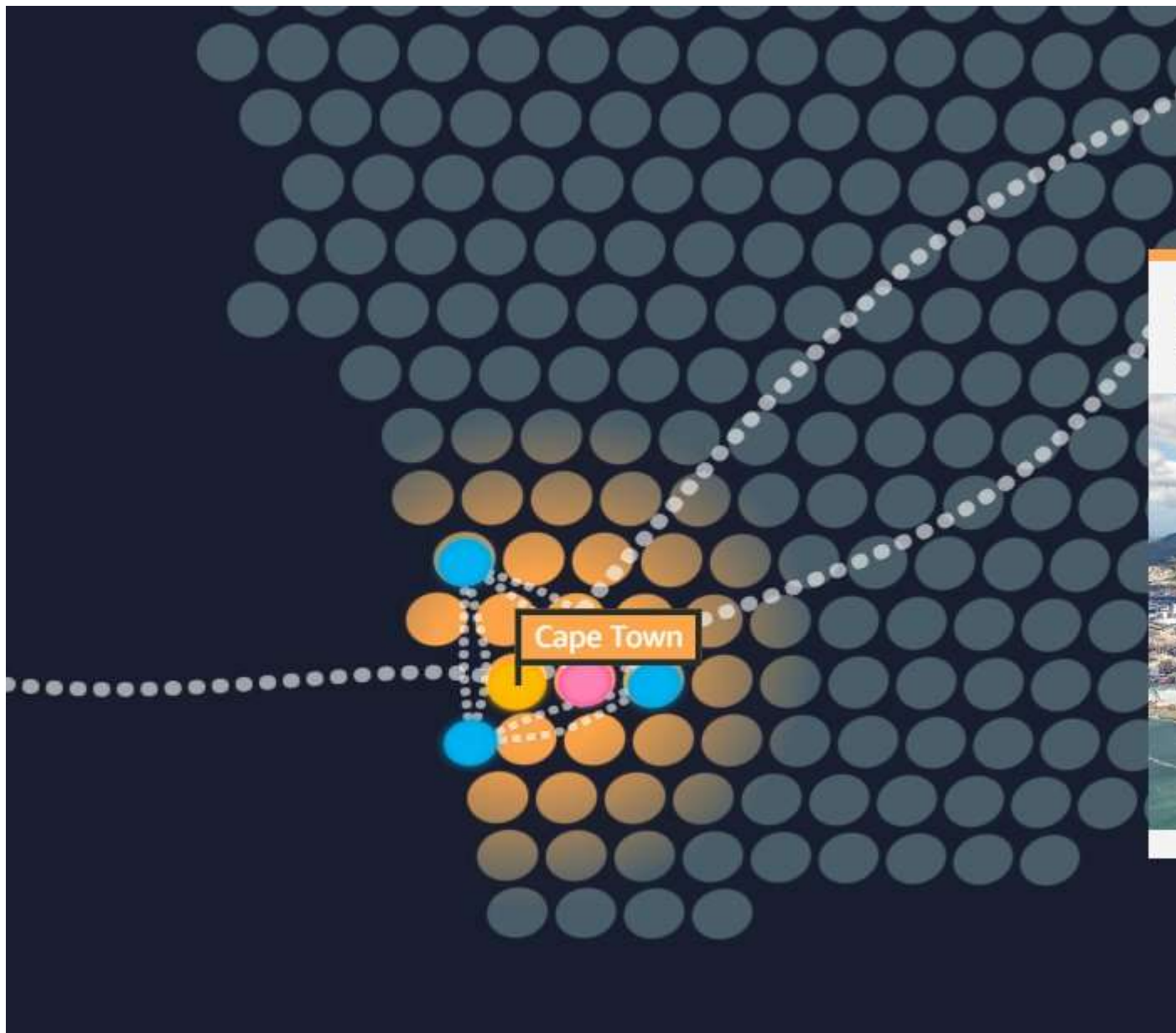
AWS designed infrastructure to solve:

- Hardware failure
- Power outages
- Natural disasters
- Traffic spikes
- Geographic latency

Cloud architecture begins with understanding where systems run.

2 Core Components of AWS Global Infrastructure

AWS Regions





A Region is:

A separate geographic area in the world where AWS clusters data centers.

Examples include:

- Africa (Cape Town)
- Europe (Frankfurt, London)
- US (N. Virginia, Ohio)
- Asia Pacific (Tokyo, Singapore)

Key Characteristics

- Completely isolated from other Regions
 - Contains multiple Availability Zones
 - Data does NOT automatically replicate between Regions
 - You choose the Region when deploying resources
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Why Region Selection Matters

1. **Latency**
 - Closer Region = faster response time
 2. **Compliance**
 - Some laws require data to remain in specific countries
 3. **Cost**
 - Pricing varies slightly by Region
 4. **Disaster Recovery**
 - Multi-Region architectures increase resilience
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Availability Zones (AZs)

Each Region contains multiple AZs.

An Availability Zone is:

- One or more physically separate data centers
 - With independent power, cooling, and networking
 - Connected by high-speed private links
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Why AZs Exist

Hardware fails.

Power systems fail.

Network equipment fails.

If your architecture uses:

- Only 1 AZ → Single point of failure
 - Multiple AZs → High availability
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Exam Rule

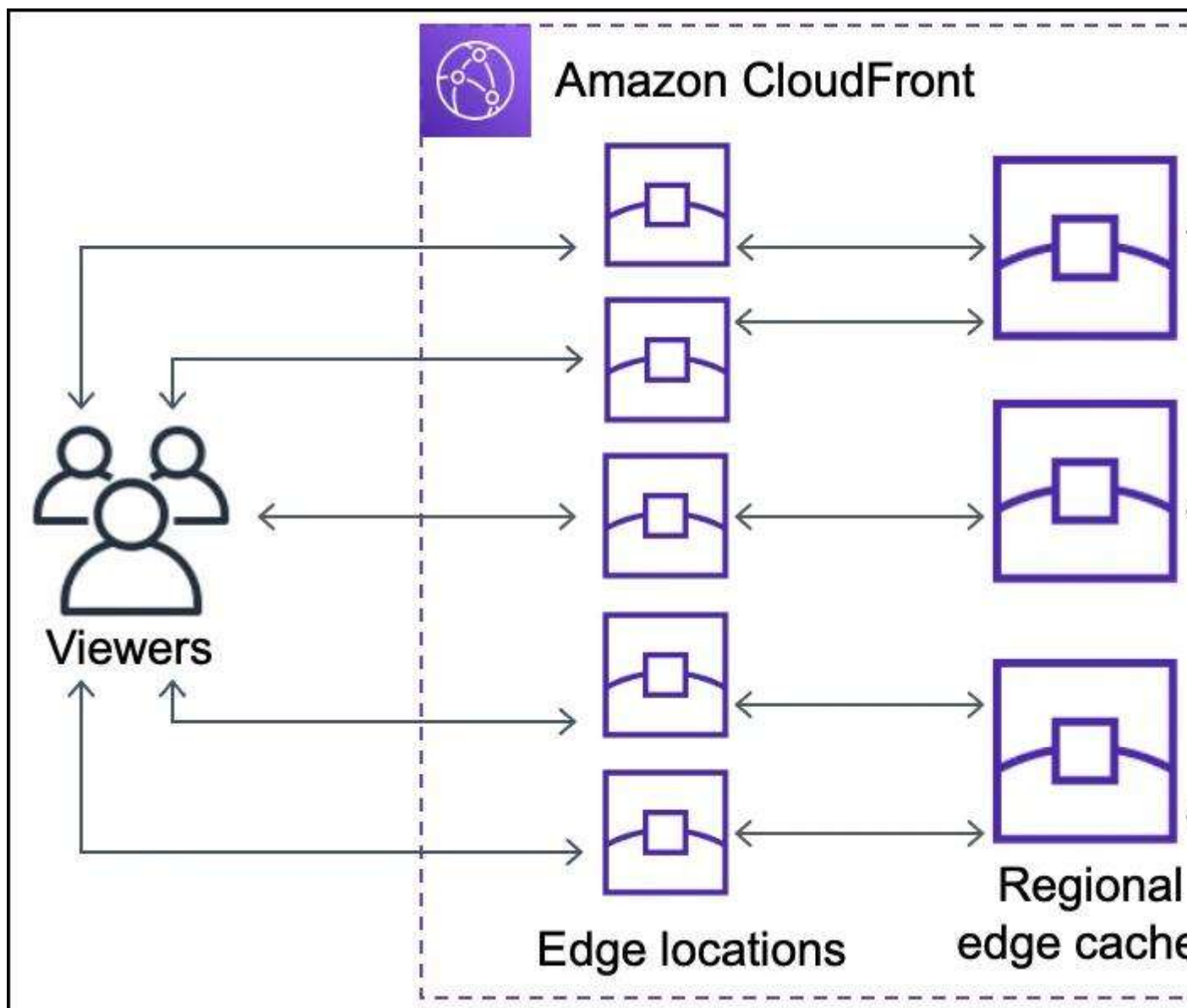
If a question says:

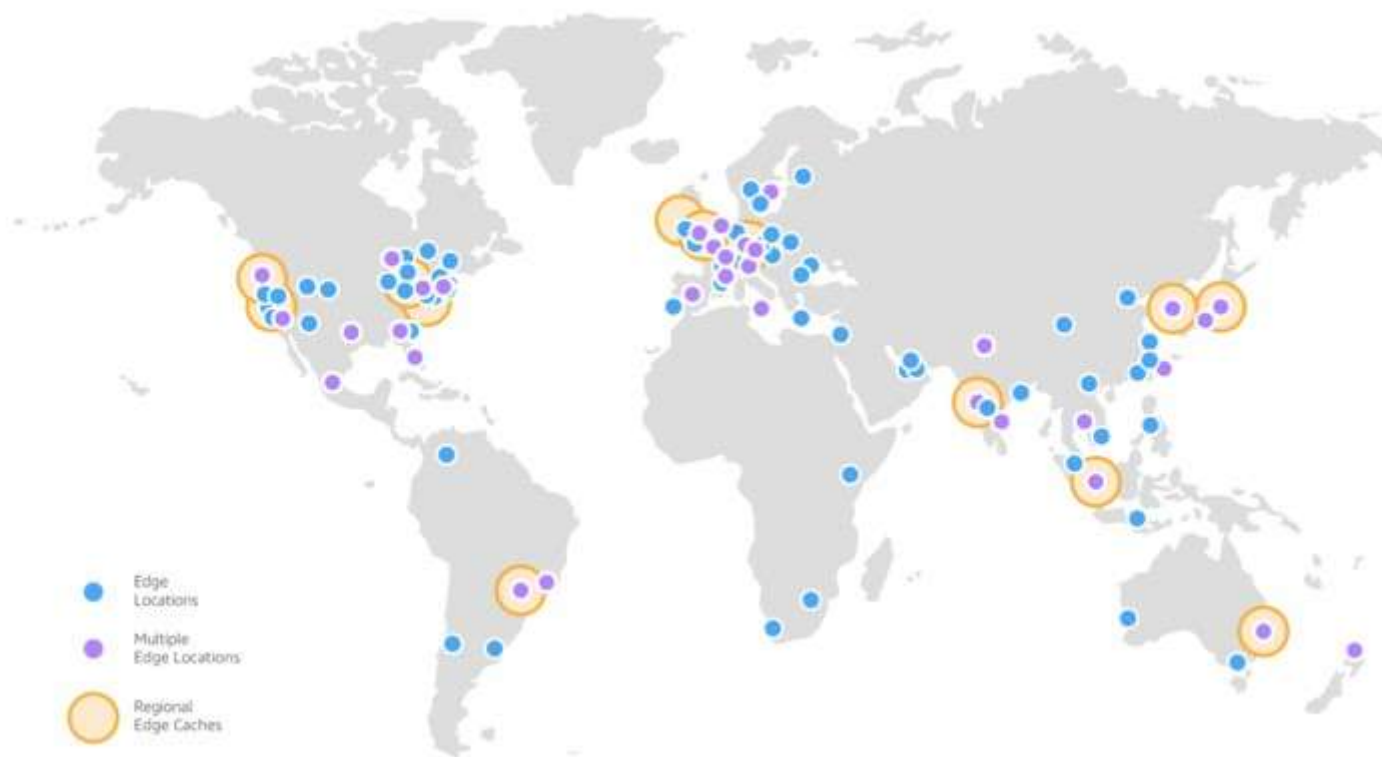
“High availability”

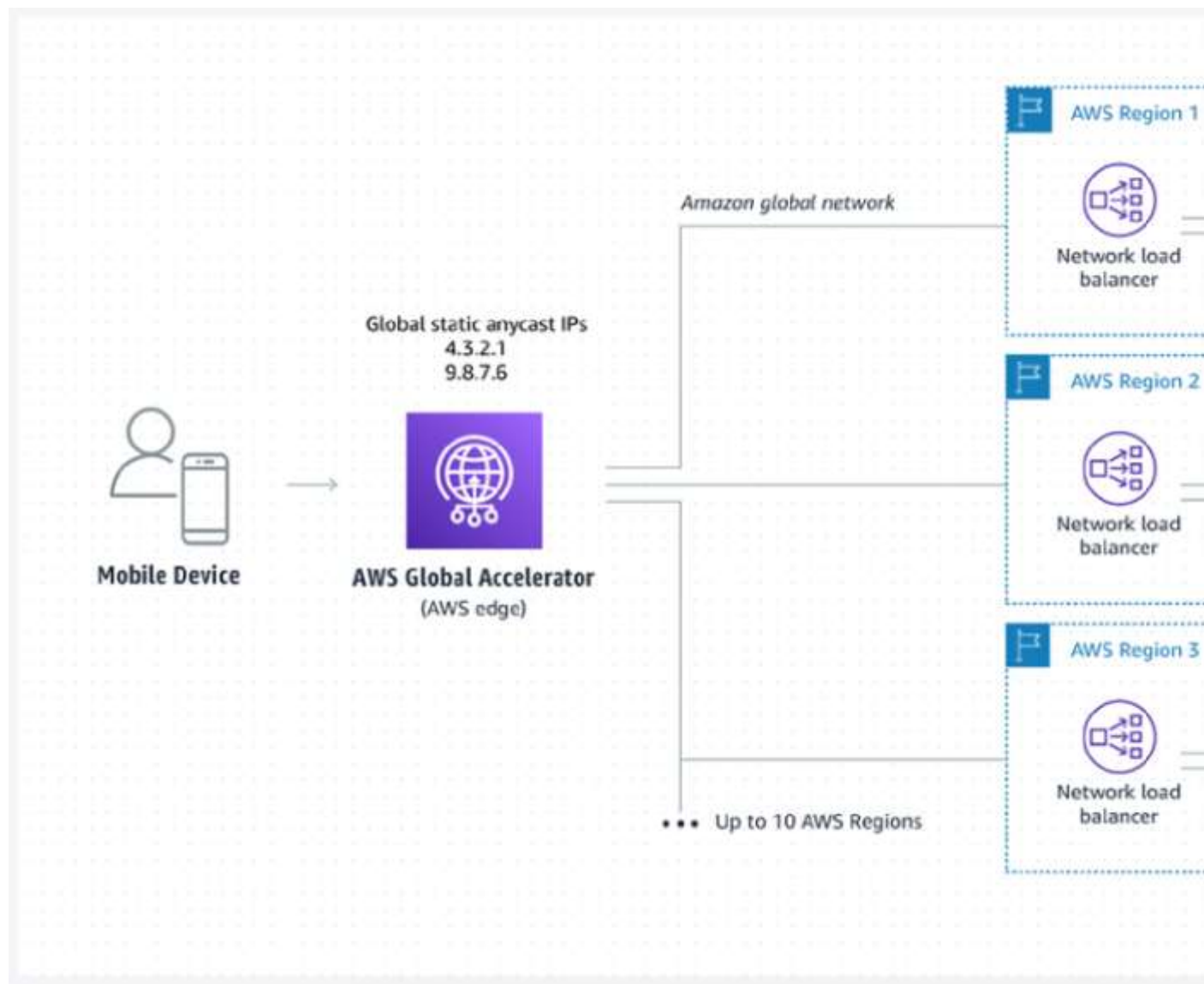
The answer almost always involves:

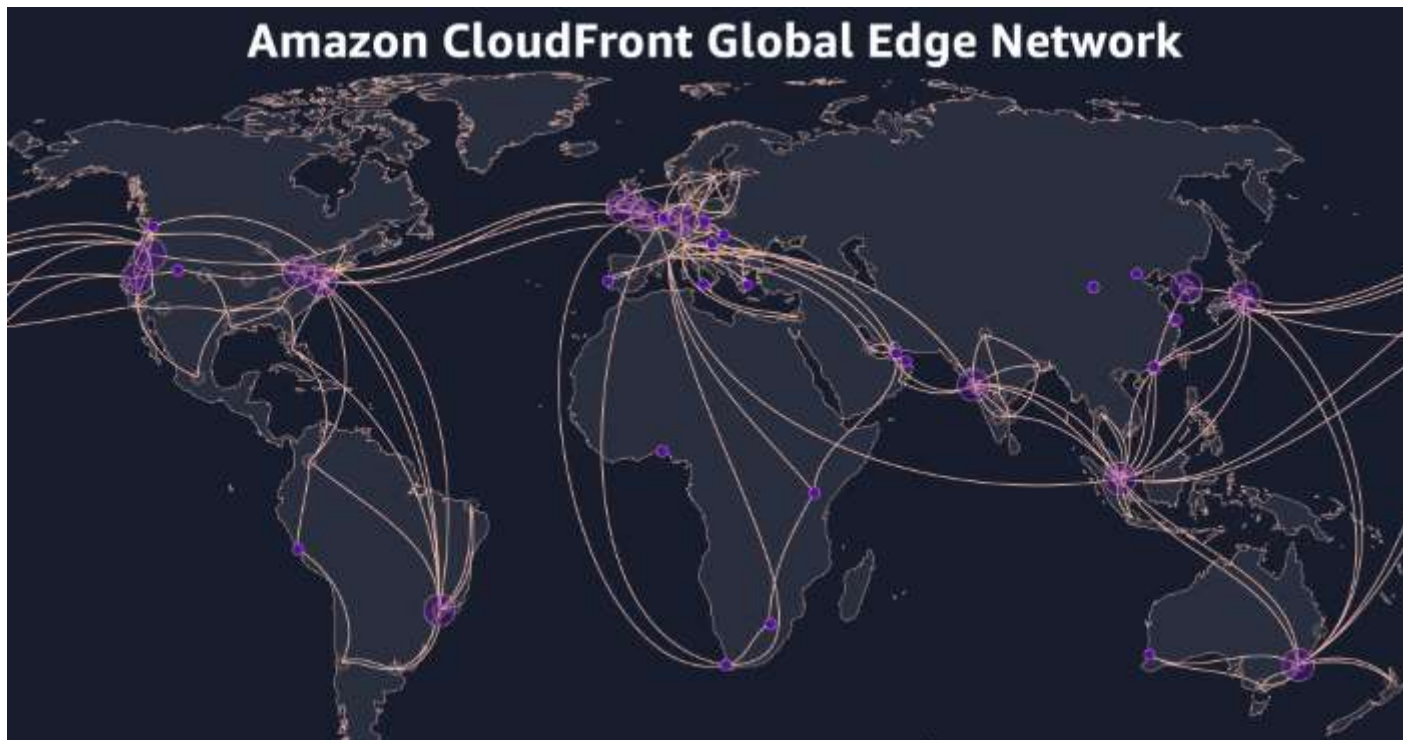
Multi-AZ deployment.

3 Edge Locations & Content Delivery









Edge Locations are:

- Smaller AWS sites distributed globally
- Used for content caching
- Reduce latency for end users

Used by:

- Amazon CloudFront (CDN)

Example:

A user in Kenya accessing a website hosted in Europe:

Without CDN → long latency

With CDN → cached locally at Edge Location

4 High Availability vs Fault Tolerance

Students must distinguish these.

High Availability

System remains operational even if one component fails.

Example:

EC2 instances across 2 AZs.

Fault Tolerance

System continues operating with no noticeable interruption.

Often requires:

- Load balancing
- Automatic failover
- Redundancy

5 Multi-AZ vs Multi-Region

Students often confuse these.

Multi-AZ	Multi-Region
Within same Region	Across different Regions
Protects against data center failure	Protects against regional disaster
Lower complexity	Higher complexity
Common design	Advanced design

Practitioner-level expectation:
Understand the difference.

6 How AWS Achieves Durability

Example: Amazon S3

S3 stores data across multiple AZs automatically.

That's how it achieves:

99.999999999% durability (11 nines).

This means:
Data loss probability is extremely low.

7 Architecture Thinking Based on Infrastructure

When designing a system, students should ask:

1. What Region is closest to users?
2. Do I need Multi-AZ?
3. Do I need Multi-Region?
4. What happens if one AZ fails?
5. What happens if one Region fails?

These are architectural questions.

8 Real Exam Triggers

Look for keywords:

- “Disaster recovery”
- “High availability”
- “Low latency”
- “Global users”
- “Compliance requirements”
- “Regional isolation”

These usually test understanding of:
Region, AZ, Edge Locations.

9 Common Exam Mistakes

- ✗ Assuming data replicates automatically across Regions
 - ✗ Thinking one AZ is sufficient for high availability
 - ✗ Confusing Edge Location with Region
 - ✗ Ignoring latency considerations
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10 Practical Student Exercise

Ask students to design:

A simple web application for users in Africa and Europe.

They must decide:

- Which Region?
- Single AZ or Multi-AZ?
- Do we need CloudFront?
- What happens if one AZ fails?

If they can justify decisions clearly,
they understand the topic.

Final Lock-In Statements

Students should confidently say:

- “Regions are isolated geographic areas.”
 - “Availability Zones protect against data center failure.”
 - “Multi-AZ increases availability.”
 - “Multi-Region increases disaster recovery capability.”
 - “Edge locations reduce latency for global users.”
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Why This Topic Is Powerful

Understanding global infrastructure means:

- You understand how AWS prevents downtime.
- You understand how scaling is geographically distributed.
- You understand disaster recovery at a conceptual level.

This is foundational knowledge.
